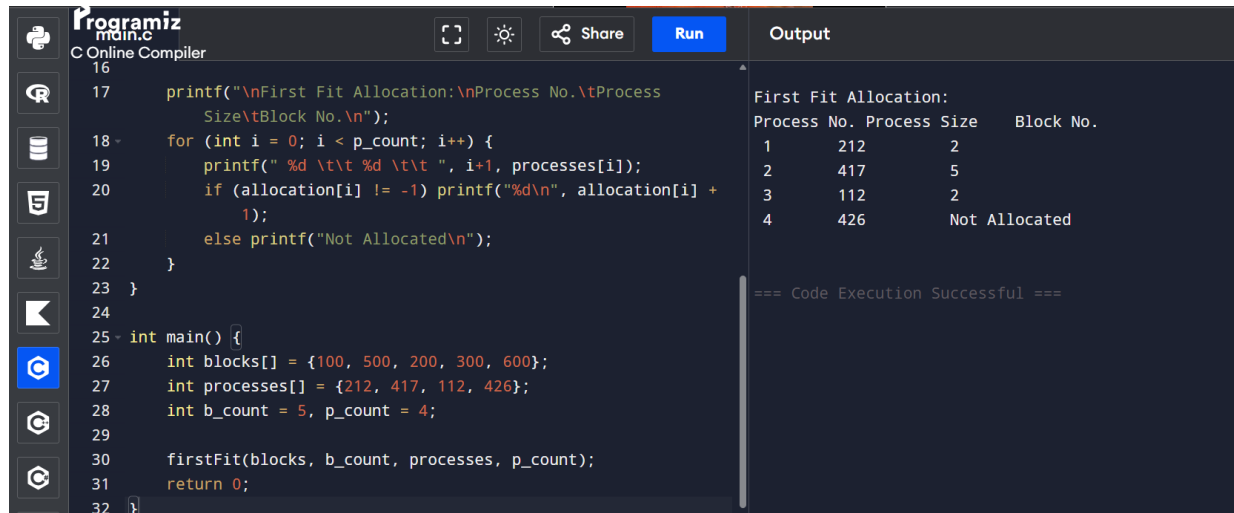


EXPERIMENT 14:

14. Construct a C program to organize the file using single level directory.



```
Programiz
main.c
C Online Compiler
16
17     printf("\nFirst Fit Allocation:\nProcess No.\tProcess
        Size\tBlock No.\n");
18     for (int i = 0; i < p_count; i++) {
19         printf(" %d \t\t %d \t\t ", i+1, processes[i]);
20         if (allocation[i] != -1) printf("%d\n", allocation[i] +
            1);
21         else printf("Not Allocated\n");
22     }
23 }
24
25 int main() {
26     int blocks[] = {100, 500, 200, 300, 600};
27     int processes[] = {212, 417, 112, 426};
28     int b_count = 5, p_count = 4;
29
30     firstFit(blocks, b_count, processes, p_count);
31     return 0;
32 }
```

Output

First Fit Allocation:

Process No.	Process Size	Block No.
1	212	2
2	417	5
3	112	2
4	426	Not Allocated

=== Code Execution Successful ===

PSEUDO CODE :

1. Define a directory structure with directory name, file count, and file list array.
2. Loop interactive menu:
 - Option 1 (Create File): Check for unique name, add to list.
 - Option 2 (Delete File): Search list, remove and shift entries.
 - Option 3 (Search File): Check if file exists in list.
 - Option 4 (Display Files): Loop and print all files in directory

CODE:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
struct Directory {
```

```
    char dname[10];
```

```
    char fname[10][10];
```

```
    int fcount;
```

```
} dir;
```

```
int main() {
```

```
    dir.fcount = 0;
```

```
    strcpy(dir.dname, "ROOT");
```

```
    int choice;
```

```
    char f[10];
```

```
    while(1) {
```

```
        printf("\n1. Create File  2. Delete File  3. Display Files  4. Exit\nEnter Choice: ");
```

```
        scanf("%d", &choice);
```

```
        if(choice == 1) {
```

```
            printf("Enter File Name: ");
```

```
            scanf("%s", dir.fname[dir.fcount]);
```

```
            dir.fcount++;
```

```
            printf("File Created.\n");
```

```
        }
```

```
        else if(choice == 2) {
```

```
            printf("Enter File Name to Delete: ");
```

```
            scanf("%s", f);
```

```
int found = 0;

for(int i = 0; i < dir.fcount; i++) {

    if(strcmp(f, dir.fname[i]) == 0) {

        printf("File %s Deleted.\n", f);

        strcpy(dir.fname[i], dir.fname[dir.fcount - 1]);

        dir.fcount--;

        found = 1;

        break;

    }

}

if(!found) printf("File Not Found.\n");

}

else if(choice == 3) {

    printf("\nDirectory: %s\nFiles: ", dir.dname);

    for(int i = 0; i < dir.fcount; i++) printf("%s ", dir.fname[i]);

    printf("\n");

}

else break;

}

return 0;

}
```